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A Data-Driven Approach to Symptom-Based Disease Prediction and Generic Medicine Suggestion

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Introduction

- Decisions begin with patient symptoms.
- Systems enable early detection, safe advice.
- This research proposes a clinical decision-support system.
- It maps symptoms to diseases and recommended generics/medicines
- For early guidance; no diagnosis/prescription.

Problem Statement

- Symptoms are unstructured and informal.
- Mapping to disease and its generic (chemical compound) is challenging.
- Existing systems ignore generic mapping.
- NLP-based automation is needed.

Motivation

- Symptom Focus: Primarily predict diseases from symptoms.
- Med Recommendations: Limited or poorly designed.
- Drug Distinctions: Ignore generic vs. brand.
- Safety Gaps: Insufficient clinical safety checks.
- Context Issues: In developing countries, generics dominate; brands vary by market.

Objective

- To map symptoms to diseases and medicines through a generic-first pipeline.
- To recommend generics and link to available brand medicines.
- To ensure guidance is explainable and clinically safe.
- To validate the system using large-scale structured datasets.

Literature Review

No.	Name of the Paper	Authors	Publication & Year	Key Findings	Research Gap
01	*End-to-End Personalized Medical Recommendation Framework for Disease Prediction*	Justin M. George, Emmanuel Joy, Suganthi Evangeline	IEEE ICSADL, 2025	Uses **BERT + LSTM** for disease and medicine prediction • Trained on **Kaggle/MIMIC** datasets • Achieves **96% & 91% disease accuracy**, **82% medicine accuracy** • Uses **133 symptom features** • Provides disease details, medicines, diet, exercise, precautions	• Limited support for rare diseases • Poor handling of incomplete/unstructured data • Cannot handle multiple diseases • No real-time EHR or wearable integration
02	*Medicine Recommendation System Using Machine-Learning Comparative Analysis|*	Abhijit Ranjan, Chandra-veer Singh	IJSRE, Sept–Oct 2025	• Compares **NB, DT, SVM, RF** models • Uses structured CSV symptom–disease data • Accuracy ranges **89.2%–95%** • Covers **41 diseases** • Provides medicine, diet, exercise advice	• Dataset is overly clean • Low robustness to noisy inputs • No dosage or drug-interaction support • No clinical validation

Literature Review

No.	Name of the Paper	Authors	Publication & Year	Key Findings	Research Gap
03	*Disease Prediction Based on Symptoms and Drug Recommendation*	Apoorva Jindal et al.	IEEE ICRITO, 2024	• Disease prediction using **RF, NB, LR, DT** • Disease accuracy up to **98%** • Drug recommendation accuracy only **38–39%** • Includes dashboards and scheduling	• Extremely small dataset (41 samples) • Poor drug recommendation performance • No personalization (history/allergies) • No cross-validation
04	*A Comprehensive Approach for Healthcare Decision-Making Through Integrated Data Mining and NLP-Enhanced Drug Recommendation Systems*	Kartheek Garapati et al.	IEEE IC-EC Tech, 2024	• Combines disease prediction with **NLP-based drug recommendation** • Uses sentiment-aware drug reviews • Achieves up to **98.5% accuracy (TF-IDF + SVM)**	• No real-time or clinical validation • Limited disease coverage • No dosage, interaction, or patient profiling • Lacks explainability

Literature Review

No.	Name of the Paper	Authors	Publication & Year	Key Findings	Research Gap
05	*Deep Learning–Enhanced Medical System for Disease Prediction, Medication Recommendation, and Intelligent Question-Answering*	Kabiru Uba Kiru et al.	IEEE UPCON, 2024	• Uses **BERT and LSTM** models • Disease prediction accuracy up to **96%** • QA exact match score **86%** • Integrates prediction, recommendation, and QA	• Tested only on offline datasets • No patient-specific or safety factors • No explainability • Dataset bias and scalability concerns
06	*An Intelligent Disease Prediction and Drug Recommendation Prototype Using Multiple ML Approaches*	Suvendu Kumar Nayak et al.	IEEE Access, 2023	• Uses **SVM and Naive Bayes** for disease prediction • Maps symptoms to diseases effectively • Supports early clinical decision-making	• No real-world clinical validation • No medicine recommendation or dosage • No personalization or explainability

Dataset Description

Dataset-1: (Symptom_Diseases.csv, Source: Kaggle)

- Row = 246945, Column = 378
- Sample Diseases: 773 (Panic Disorder, Vaginitis, Angina, etc.)
- Sample Symptoms: 246 (Anxiety and nervousness, Depression, Insomnia, etc.)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	diseases	anxiety an	depressio	shortness	depressive	sharp che	dizziness	insomnia	abnormal	chest tight	palpitatio	irregular h	breathing	hoarse vo	sore throa	difficulty s	cough	nasal cong	throat swe di
2	panic diso	1	0	1	1	0	0	0	0	1	1	1	1	0	0	0	0	0	0
3	panic diso	0	0	1	1	0	1	1	0	0	1	0	0	0	0	0	0	0	0
4	panic diso	1	1	1	1	0	1	1	0	0	1	1	1	0	0	0	0	0	0
5	panic diso	1	0	0	1	0	1	1	1	0	0	0	1	0	0	0	0	0	0
6	panic diso	1	1	0	0	0	0	1	1	1	0	0	0	0	0	0	0	0	0
7	panic diso	0	0	1	1	0	0	1	0	0	1	1	0	0	0	0	0	0	0
8	panic diso	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
9	panic diso	0	0	0	1	0	0	1	1	1	1	1	1	0	0	0	0	0	0
10	panic diso	1	0	0	1	0	1	1	0	1	0	1	0	0	0	0	0	0	0
11	panic diso	1	1	1	0	0	0	0	1	0	1	1	1	0	0	0	0	0	0
12	panic diso	0	0	1	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0
13	panic diso	1	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0
14	panic diso	1	1	1	0	0	0	1	0	1	1	1	0	0	0	0	0	0	0
15	panic diso	1	0	0	1	0	0	0	1	1	0	0	0	0	0	0	0	0	0

Figure 1: A small portion of **Symptom_Diseases.csv** file

Dataset Description

Dataset-2: (Generic.csv, Source: Kaggle)

- Row = 1711, Column = 22
- Sample Generics: 1711 (Adalimumab, Acyclovir + Hydrocortisone, Acyclovir, Acemetacin, Aceclofenac, Fumarate, Acitretin etc.)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	generic id	generic na	slug	monograp	drug class	indication	indication	therapeut	pharmaco	dosage de	administra	interactio	contraindi	side effect	pregnancy	precautio	pediatric	overdose	duration c re
2	31	Adalimum	adalimum	https://m	Immunos	Ulcerative	Adalimu	Disease-m	Adalimum	Rheumat	Administe	Abatacept	Adalimum	The most	Pregnancy		Pediatric	The maximum tolerate	
3	10	Acyclovir +	acyclovir-h	https://m	Hydrocort	Herpes lat	Acyclovir &	Hydrocort	Acyclovir	The cream should be topically a			There is n	The follow	Pregnancy	Acyclovir a	Safety and effectiveness in pediat		
4	9	Acyclovir (	acyclovir-t	https://m	Topical An	Sore lips	AcyclovirÂ	Topical An	Acyclovir i	AcyclovirÂ cream sho		Clinical ex	AcyclovirÂ	The most	Pregnancy	AcyclovirÂ	Pediatric Use: An open-label, unco		
5	12	Acyclovir (	acyclovir-c	https://m	Herpes sir	Varicella z	Aciclovir	Herpes sir	Aciclovir is	Treatmen		Probeneci	Aciclovir is	Rash, gast	Pregnancy	Aciclovir should be administered with cautio			
6	13	Acyclovir (	acyclovir-c	https://m	Ophthalm	Neonatal (	Acyclovir i	Ophthalm	Acyclovir i	The dosage for all ag		No clinical	Acyclovir i	Very comr	Pregnancy	The recommended d		No adverse effects wou	
7	7	Acyclovir (	acyclovir-i	https://m	Herpes sir	Varicella z	Acyclovir	Herpes sir	Acyclovir e		It is	Co-admini	Acyclovir I	Some infr	Pregnancy	Acyclovir I	Pediatric	Overdosa	It is recom Ea
8	22	Activated	activated-	https://m	Antidote p	Poisoning	Diarrhoea	Anti-diarrl	Active cha	Diarrhoe		Ultracarbc	Febrile dia	As most u	No data fc	Several poisons and		Activated charcoal is w	
9	2060	Aclidinium	aclidinium-bromide-		Combined	bronchod	It is indica	Combined	Aclidinium	For oral inhalation o		Use of		Most com	There are				
10	25	Acitretin	acitretin-2	https://m	Oral Retin	Scaly skin	Acitretin is	Oral Retin	Acitretin is	Adult and elderly: Ini		Existing da	Acitretin is	Adverse e	Acitretin is	The risk of giving birt		Manifestations of acute	
11	177	Acidic com	acidic-component-h		Haemodia	Haemodia	This soluti	Haemodia	Calcium ch	Recommended dose		There are	Known hy	Fluid and	Pregnancy	It is used for hemodialysis purposes only. Do			
12	6	Acetylcyst	acetylcyst	https://m	Antidote p	Tuberculo	Acetylcyst	Antidote p	Acetylcyst	Effervesc		The thiol g	Known hy	Like all me	In case of	Patients suffering fro		Overdose of Acetylcyst	
13	26	Acetazolar	acetazolar	https://m	Carbonic a	Tumor lys	Acetazolar	Carbonic a	Acetazolar	Adults : Initially 250 mg (1 table			Acetazolar	Common :	There are no adequate and well controlled studies in p				
14	184	Acetate Fo	acetate-formulation-		Haemodia	Haemodia	This soluti	Haemodia	Calcium ch	Recommended dose		There are	Known hy	Fluid and	Pregnancy	It is used for haemodialysis purposes only.			
15	21	Acemetaci	acemetaci	https://m	Non-stero	Spondyliti	Acemetac	Non-stero	Acemetaci	The recommended s		Acemetaci	Acemetaci	Common :	The safety	Caution should be taken in elderly people, h			
16	3	Aceclofen	aceclofen	https://m	Drugs for	Spondyliti	Aceclofen	Drugs for	Aceclofen	Extended release tal		No	Aceclofen	Aceclofen	The use o	Caution sh	There are no clinical data on the u		

Figure 2: A small portion of **Generic.csv** file

Dataset Description

Dataset-3: (Medicine.csv, Source: Kaggle)

- Row = 21714, Column = 10
- Sample Medicine: 5084 (A-Cold, A-Clox, A-Care, A-Card, Alovera, Alkuli, Alisa, Aliksir, Alfa-E etc.)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	brand id	brand nan	type	slug	dosage fo	generic	strength	manufactu	package c	Package Size				
2	4077	A-Cold	allopathic	a-coldsyru	Syrup	Bromhexi	4 mg/5 ml	ACME Lab	100 ml bottle: à§³ 40.12					
3	4006	A-Cof	allopathic	a-cofsyrup	Syrup	Dextrome	(10 mg+30	ACME Lab	100 ml bottle: à§³ 100.00					
4	6174	A-Clox	allopathic	a-cloxinjec	Injection	Cloxacillin	500 mg/vi	ACME Lab	500 mg via	(5's pack: à§³ 142.15)				
5	6173	A-Clox	allopathic	a-cloxinjec	Injection	Cloxacillin	250 mg/vi	ACME Lab	250 mg via	(5's pack: à§³ 100.00)				
6	6172	A-Clox	allopathic	a-cloxpow	Powder fo	Cloxacillin	125 mg/5	ACME Lab	100 ml bottle: à§³ 45.00					
7	6171	A-Clox	allopathic	a-cloxcaps	Capsule	Cloxacillin	500 mg	ACME Lab	Unit Price:	(100's pack: à§³ 598.00)				
8	14006	A-Care	allopathic	a-caretabl	Tablet	Betacarote	6 mg+200	Asiatic Lab	Unit Price:	(20's pack: à§³ 50.00)				
9	3039	A-Card	allopathic	a-cardtabl	Tablet	Isosorbide	20 mg	ACME Lab	Unit Price:	(100's pack: à§³ 143.00)				
10	15483	A-Calm	allopathic	a-calmtab	Tablet	Tolperisor	50 mg	ACME Lab	Unit Price:	(50's pack: à§³ 300.00)				
11	14480	A-Cal DX	allopathic	a-cal-dxta	Tablet	Calcium C	500 mg+40	ACME Lab	Unit Price:	(30's pack: à§³ 225.00)				
12	14399	A-Cal D	allopathic	a-cal-dtab	Tablet	Calcium C	500 mg+20	ACME Lab	Unit Price:	(15's pack: à§³ 97.50),(30's pack: à§³ 195.00)				
13	14264	A-Cal	allopathic	a-caltablet	Tablet	Calcium C	500 mg	ACME Lab	Unit Price:	(100's pack: à§³ 500.00)				
14	13582	A-B1	allopathic	a-b1tablet	Tablet	Thiamine	100 mg	ACME Lab	Unit Price:	(100's pack: à§³ 86.00)				
15	25154	5X	allopathic	5xtablet30	Tablet	Ulipristal A	30 mg	Renata Lin	Unit Price: à§³ 200.00					
16	105	5-Fluril	allopathic	5-fluriliv-in	IV Injectio	Fluoroura	25 mg/ml	Techno Dr	250 mg vial: à§³ 55.00,500 mg vial: à§³ 95.00					
17	27661	Arubin	herbal	arubincap	Capsule	Nabayas L	500 mg	Square Ph	Unit Price:	(30's pack: à§³ 120.00)				
18	29715	Arkid	herbal	arkidsyrup	Syrup	Aravindasav		ACME Lab	100 ml bottle: à§³ 70.00					
19	29714	Arjunacar	herbal	arjunacar	Syrup	Arjunarista		ACME Lab	200 ml bottle: à§³ 90.00					
20	30450	Apelin	herbal	apelinsyru	Syrup	Malus sylvestris		Hamdard	100 ml bottle: à§³ 60.00,450 ml bottle: à§³ 200.00					
21	29646	Antiligo	herbal	antiligocap	Capsule	Barsina		Hamdard	Unit Price:	(50's pack: à§³ 150.00)				

Figure 3: A small portion of **Medicine.csv** file

Methodology

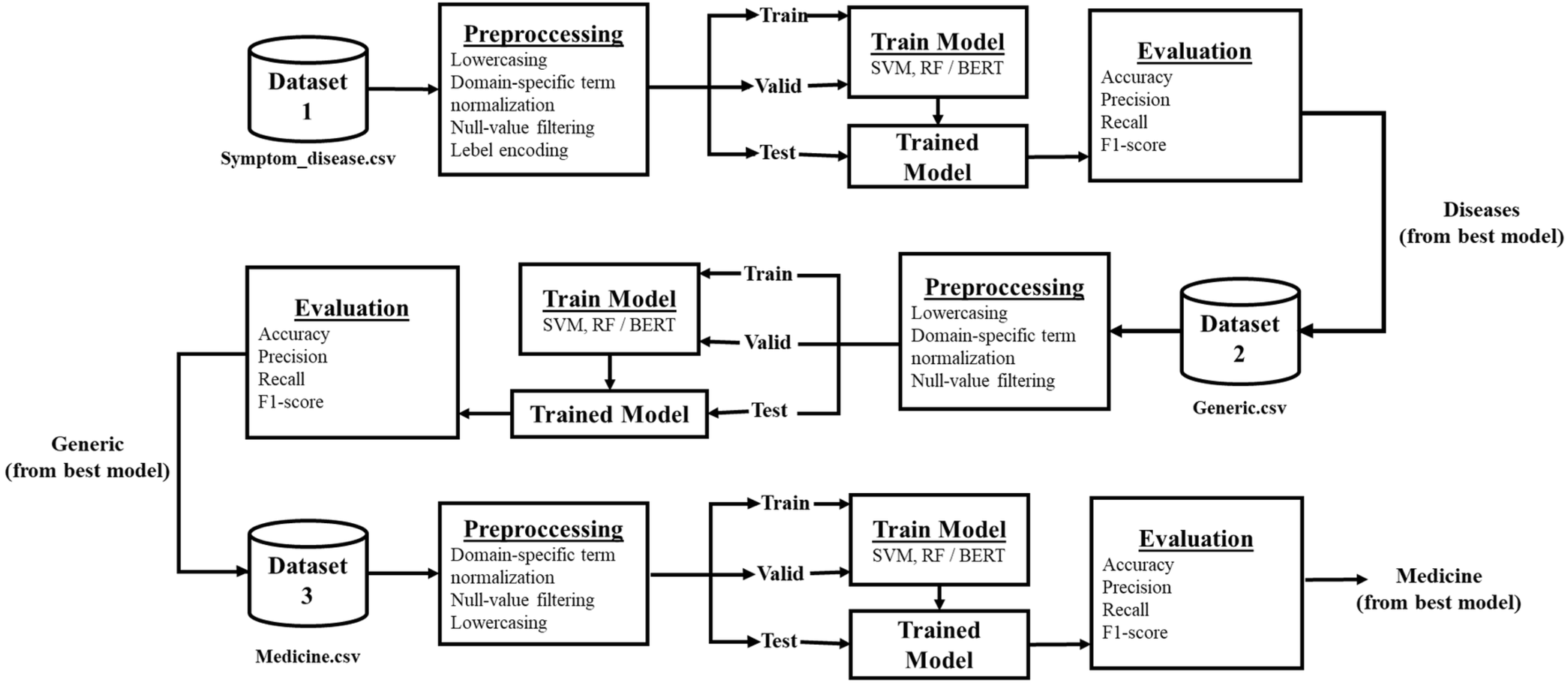

Figure 4: Proposed Methodology of the system

Required Tools

Programing Platform

- Google Colab
- VS Code
- Kaggle

Programming Language

- Python 3.14

Conclusion

In conclusion, this research develops a system that predicts diseases from patient symptoms and recommends suitable generic and brand medicines. By integrating symptom–disease data with generic and brand medicine information, the system provides a structured and safe decision-support process. This work helps improve patient awareness, supports healthcare providers, and increases access to medical knowledge in a data-driven way.

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Thank You